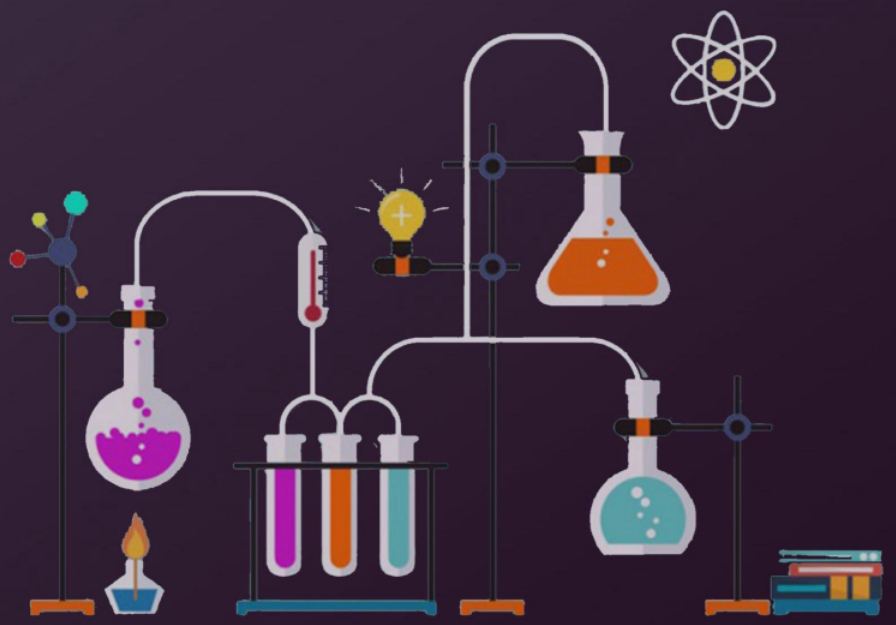
CHEMISTRY

Hydrocarbon



SHEET

eSaral.com

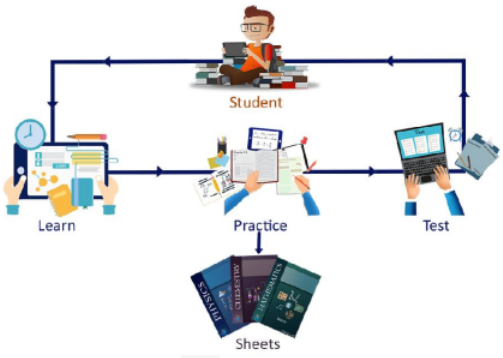
Contents

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

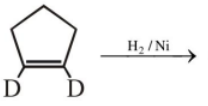
Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Preparation Methods of Alkanes : Hydrogenation and Wurtz Reaction	1-2	1	1	
2	Preparation Methods of Alkanes: Decarboxylation and Kolbe's Reaction	3-7	2	2-8	
3	Physical Properties of Alkanes				
4	Chemical Properties of Alkanes	8-16	5	9-12	
5	Preparation of Alkenes: Dehydration of Alcohol by elimination	17-19		13-15	
6	Carbocation Rearrangement				
7	Preparation of Alkenes: Dehydrohalogenation	20-21	6	16	
8	Alkenes: Other Methods of Preparation				
9	Physical Properties of Alkenes				
10	Electrophylic Addition reactions on Alkene	22-25	7	17-19	
11	Addition of Carbene, Ozonolysis and Oxidation by Bayer's Reagent	26-27		20-22	
12	Oxidation, Free Radical Substitution and polymerisation of Alkene	28		23	9
13	Conjugated Diene, KCP vs TCP				
14	Preparation: Alkyne and Physical Properties of Alkyne	29-30		24	
15	Chemical Properties of Alkyne	31-36		25-28	
16	Nucleophilic Addition Reaction and Oxidation of Alkyne				
17	Substitution and Cyclic Polymerisation of Alkyne				
18	Introduction of Aromatic Hydrocarbons				
19	Electrophilic Substitution Reactions and Oxidation of Aromatic Hydrocarbons				
20	ACC	37-40	3-4, 8	29-35	1-8

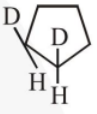
Exercise - 1 — Concept Builder

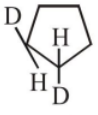
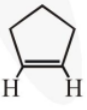
SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Preparation Methods of Alkanes : Hydrogenation and Wurtz Reaction

Q1.  A. A is :-

(A) $CH_3–(CH_2)_4–CH_3$ (B) 

(C)  (D) 

Q2. Which of the following hydrocarbon cannot be obtained by sabatier and Senderen's reaction :-

(A) CH_4 (B) C_2H_6

(C) C_3H_8 (D) All

Preparation Methods of Alkanes: Decarboxylation and Kolbe's Reaction

Q3. Anhydrous solution of the following compound on electrolysis gives ethane :-

(A) Acetic acid

(B) Acetamide

(C) Potassium acetate

(D) Ethyl acetate

Q4. Sodium propionate on decarboxylation with sodalime gives :-

(A) Propane (B) Ethane

(C) Butane (D) Pentane

Q5. Which of the following alkanes cannot be produced by Kolbe electrolysis of sodium or potassium salts of carboxylic acids ?

(A) Methane (B) Ethane

(C) Butane (D) Hexane

Q6. Decarboxylation of isobutyric acid yields :-

(A) Isobutane

(B) Propane

(C) 2-Methylpropane

(D) None of the above

Q7. Rate of decarboxylation of following carboxylic acid with sodalime will be in order.

$CH_3COOH \xrightarrow[r_1]{NaOH+CaO}$

$CH_3CH_2COOH \xrightarrow[r_2]{NaOH+CaO}$

$CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–COOH \xrightarrow[r_3]{NaOH+CaO}$

(A) $r_1 > r_2 > r_3$ (B) $r_3 > r_2 > r_1$

(C) $r_2 > r_3 > r_1$ (D) $r_1 > r_3 > r_2$

Chemical Properties of Alkanes

Q8. In the chlorination of methane, one of the chain propagating step is :-

(A) $CH_3\bullet + Cl\bullet \rightarrow CH_3Cl$

(B) $CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$

(A) $CH_3\bullet + CH_3 \rightarrow CH_3–CH_3$

(D) $Cl\bullet + Cl\bullet \rightarrow Cl_2$

Q9. The homolytic fission of hydrocarbon results in the formation of -

(A) Free radicals

(B) Carbocations

(C) Carbanions

(D) Carbenes

Q10. n-Heptane when heated to a temperature of about 800 K under high pressure in the presence of Cr_2O_3/Al_2O_3 catalyst gives -

(A) 1-heptene (B) 2-Methylhexane
(C) Toluene (D) Xylene

Q11. Number of structural isomer which can be theoretically obtained on monochlorination of 2-methylbutane is -

(A) 6 (B) 2 (C) 3 (D) 4

Q12. The least reactive alkane towards free-radical substitution reactions is -

(A) CH_4 (B) $(CH_3)_3CH$
(C) CH_3CH_3 (D) $CH_3CH_2CH_3$

Q13. Arrange the following in correct order of reactivity towards Cl_2/hv -

(a) CH_4 (b) CH_3CH_3
(c) $CH_3CH_2CH_3$ (d) $CH_3-\underset{CH_3}{\overset{|}{C}}H-CH_3$

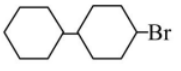
(A) a > b > c > d (B) d > c > b > a
(C) b > c > a > d (D) c > b > d > a

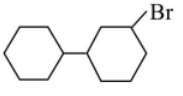
Q14. n-hexane $\xrightarrow[600°C]{Cr_2O_3/Al_2O_3}$ Cyclic compound.
The wrong title of the reaction would be :-

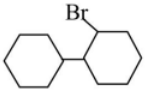
(A) Aromatisation (B) Hydroformation
(C) Cyclisation (D) Isomerisation

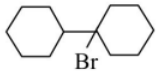
Q15. Major product for the reaction is -

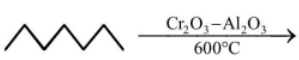


(A) 

(B) 

(C) 

(D) 

Q16.  Major product :

(A) 
(B) 
(C) 
(D) 

Preparation of Alkenes: Dehydration of Alcohol by elimination

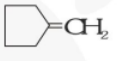
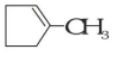
Q17. Alkenes can be prepared by :-

(A) Dehydration of alcohols
(B) Kolbe's electrolytic synthesis
(C) Dehydrohalogenation
(D) All of these

Q18. For the reaction



the major product is :

(A) 
(B) 
(C) 
(D) 

Q19. The order of reactivity of alcohols with respect to acid catalysed elimination will be:

(a) $(CH_3)_3C-OH$
(b) $CH_3CH_2CH_2CH_2OH$
(c) $CH_3\underset{OH}{\overset{|}{C}}H-CH_2-CH_3$

Code is :-

(A) a > b > c (B) a > c > b
(C) b > a > c (D) b > c > a

Preparation of Alkenes: Dehydrohalogenation

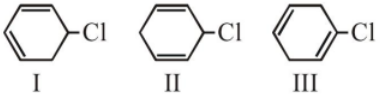
Q20. An halide $C_5H_{11}Br$ on treatment with alc. KOH give 2–pentene only. The halide will be :-

(A) $CH_3–CH_2–CH_2–CH_2–CH_2–Br$

(B) $CH_3–CH_2–CH_2–\underset{\displaystyle Br}{\underset{|}{CH}}–CH_3$

(C) $CH_3–CH_2–\underset{\displaystyle Br}{\underset{|}{CH}}–CH_2–CH_3$

(D) $CH_3–\overset{\displaystyle CH_3}{\overset{|}{CH}}–\underset{\displaystyle Br}{\underset{|}{C}}–CH_3$

Q21. 

I II III

Which one is most easily dehydrohalogenated ?

(A) I

(B) II

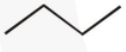
(C) III

(D) all with same case

Electrophylic Addition Reactions On Alkene

Q22. Which of the following will give positive test of unsaturation with Bromine water -

(A)  (B) 

(C)  (D) 

Q23. Which one of the following will not give positive test with bromine water ?

(A)  (B) 

(C) $CH_2 = CH_2$ (D) $CH \equiv CH$

Q24. The test for unsaturation is confirmed by the decolourisation of which of the following :-

(A) Iodine water (B) $CuSO_4$ solution

(C) Bromine water (D) All of these

Q25. Two jars A and B are filled with hydrocarbons. Br_2 in CCl_4 is added to these jars. A does not decolourise the Br_2 solution, but B decolourises. What are A and B respectively :

(A) Alkane and alkene

(B) Alkene and alkane

(C) Both

(D) None

Addition of Carbene, Ozonolysis and Oxidation by Beayer's Reagent

Q26. Dilute aqueous $KMnO_4$, at room temperature reacts with R – CH = CH – R to give :-

(A) R – CHO

(B) R – COOH

(C) RCHOH – CHOHR

(D) $CO_2 + H_2O$

Q27. In the following sequence of reactions, the alkene affords the compound 'B'

$CH_3\ CH = CHCH_3 \xrightarrow{O_3} A \xrightarrow[Zn]{H_2O} B.$

The compound B is :-

(A) CH_3COCH_3

(B) $CH_3CH_2COCH_3$

(C) CH_3CHO

(D) CH_3CH_2CHO

Oxidation, Free Radical Substitution and polymerisation of Alkene

Q28. Propene when heated with chlorine at about 500° C forms :-

(A) $CH_2\ Cl.CH = CH_2$

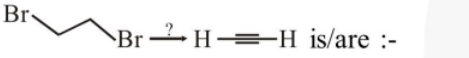
(B) $CH_3.\ CHCl.CH_2Cl$

(C) $CH_2Cl.CHCl.CH_2Cl$

(D) All the three

Preparation: Alkyne and Physical Properties of Alkyne

Q29. $R—CH_2—CCl_2—R \xrightarrow{\text{Reagent}} R—C\equiv C—R.$
The reagent is -
(A) Na
(B) HCl in H_2O
(C) KOH in C_2H_5OH followed by
(D) Zn in alcohol sodamian/ Δ

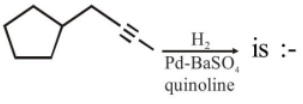
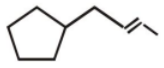
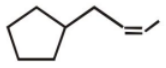
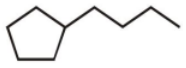
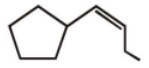
Q30. The reagent(s) for the following conversion,

is/are :-
(A) Alcoholic KOH
(B) Alcoholic KOH followed by $NaNH_2$
(C) aqueous KOH followed by $NaNH_2$
(D) Zn/CH_3OH

Chemical Properties of Alkyne

Q31. Propyne on reaction with aqueous chlorine gives -
(A) 1, 1, 2, 2-Tetrachloropropane
(B) 1, 2-Dichloropropene
(C) 1, 1-Dichloropropanone
(D) 2, 2-Dichloropropanone

Q32. Mesitylene can be obtained by polymerization of -
(A) Ethyne (B) Ethene
(C) Propene (D) Propyne

Q33. $\underset{\text{2-butyne}}{CH_3-C\equiv C-CH_3} \xrightarrow[(\text{liq.})NH_3]{Na} X$
Product x is -
(A) Cis-2-butene (B) Trans-2-butene
(C) Syn-2-butene (D) E-2-butene

Q34. The product formed in the reaction

is :-
(A) 
(B) 
(C) 
(D) 

Q35. Acetylene gives :-
(A) White precipitate with $AgNO_3$ and red precipitate with Cu_2Cl_2
(B) White precipitate with Cu_2Cl_2 and red precipitate with $AgNO_3$
(C) White precipitate with both the reagents
(D) Red precipitate with both the reagents

Q36. [A] $\xleftarrow[\text{catalyst}]{\text{lindlar's}} CH_3-C\equiv C-CH_3 \xrightarrow[\text{liq. } NH_3]{\text{Na in}}$
[B] [A] and [B] are respectively :-
(A) cis, trans-2-butene
(B) Both trans-2-butene
(C) trans cis-2- butene
(D) Both cis -2- butene

After Complete Chapter

Q37. Which of the following reactions can be employed for getting odd carbon alkanes in good yield ?
(A) Wurtz reaction
(B) Corey–House reaction
(C) Both
(D) None of these

Q38. The ease of formation of free radicals follows the order -
(A) $3^\circ > 2^\circ > 1^\circ > \dot{C}H_3$
(B) $\dot{C}H_3 > 1^\circ > 2^\circ > 3^\circ$
(C) $1^\circ > 2^\circ > 3^\circ > \dot{C}H_3$
(D) $2^\circ > 1^\circ > 3^\circ > \dot{C}H_3$

Q39. Lindlar's catalyst is :-

(A) Pt in ethanol

(B) Pd + $CaCo_3$

(C) Ni in ethanol

(D) Na in liquid NH_3

Q40. The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives predominantly :-

(A) benzoyl chloride

(B) benzyl chloride

(C) o and p-chlorotoluene

(D) m-chlorotoluene

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A mixture of ethyl iodide and n-propyl iodide is subjected to Wurtz reaction. The hydrocarbon that are formed is-

(A) n-Butane

(B) Propyne

(C) n-Pentane

(D) n-Hexane

Q2. Aqueous solution of sodium acetate and sodium propionate on electrolysis yields :-

(A) Ethane

(B) Butane

(C) Propane

(D) Pentane

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

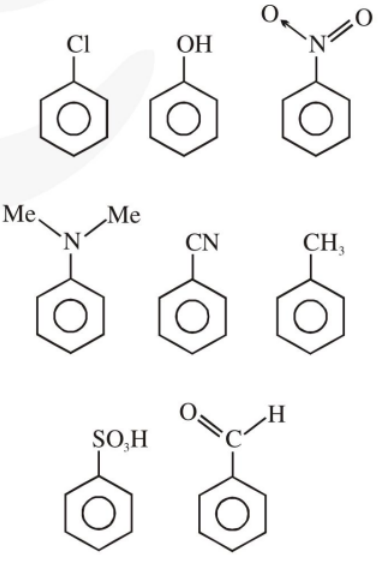
Q3. Match the column I with column II.

Column-I		Column-II	
(A)	Wurtz reaction	(p)	Electrophilic substitution reaction
(B)	Hydration of alkenes	(q)	Free radical substitution
(C)	Nitration of alkane	(r)	Electrophilic addition reaction
(D)	Reaction of alkene with NBS	(s)	Nucleophilic substitution

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q4. Number of compounds which have more activating power than benzene.



ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

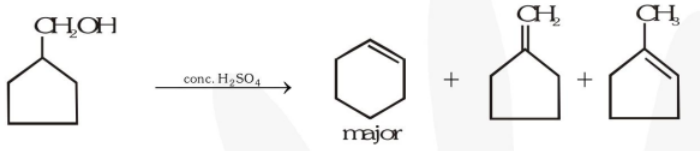
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q5. **Statement-I :** Neopentane forms only one monochlorinated product.

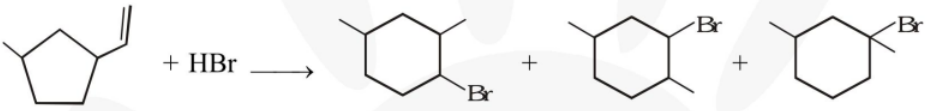
Statement-II : Neopentane has four identical methyl group attached to a quaternary carbon.

SUBJECTIVE TYPE

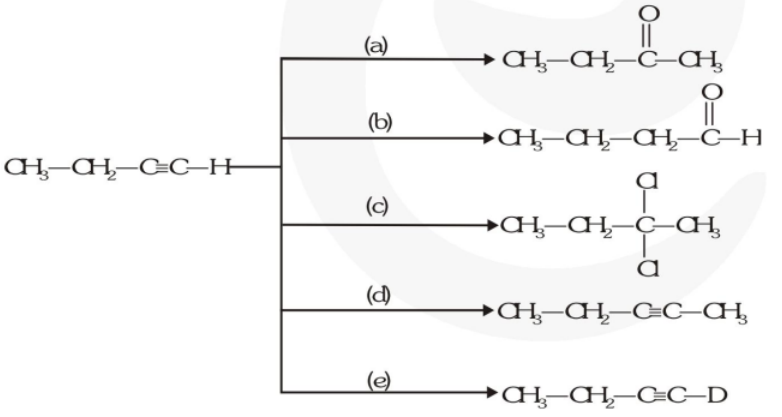
Q6. 

Give mechanism of this reaction.

Q7. Propose mechanism :



Q8. Find out reagents involved in following conversions :



SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If ethyl bromide and methyl bromide are mixed in equal proportions and the mixture is treated with sodium, predict the number of organic product formed :-

(A) 3 (B) 2

(C) 1 (D) 4

Q2. Propane will be obtained by the decarboxylation of which of the following carboxylic acid :-

(A) $CH_3CH_2CH_2COOH$

(B) $HO-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_3$

(C) $CH_3-\underset{\underset{COOH}{|}}{CH}-CH_3$

(D) All the above

Q3. The number of Structure of monocarboxylic acids which on decarboxylation form isopentane is :-

(A) 2 (B) 1

(C) 3 (D) 4

Q4. Choose the correct set showing the production of alkane by decarboxylation of the carboxylic acids :-

(A)	2,2-Dimethylpropanoic acid	(a)	isobutane
(B)	Butanoic acid	(b)	butane
(C)	2-Methylpentanoic acid	(c)	pentane
(D)	4-Methylpentanoic acid	(d)	propane

The answer is :-

(A) A and C (B) A and B

(C) B and D (D) C and D

Q5. During electrolysis which anion migrates towards anode so as to produce 2, 3-dimethyl butane :

(A) $CH_3-CH_2-COO^-$ (B) $CH_3-CH\begin{matrix}COO^-\\COO^-\end{matrix}$

(C) $CH_3-\underset{\underset{CH_3}{|}}{CH}-COO^-$ (D) CH_3COO^-

Q6. Which of the following carboxylic acids undergo decarboxylation easily -

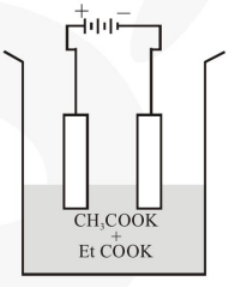
(A) $C_6H_5CO-CH_2COOH$

(B) $C_6H_5COCOOH$

(C) $C_6H_5\underset{\underset{OH}{|}}{CH_2}-COOH$

(D) $C_6H_5\underset{\underset{NH_2}{|}}{CH_2}-COOH$

Q7.



Which of the gases evolved on the surface of anode :-

(A) C_2H_6, H_2

(B) C_3H_8, H_2

(C) $C_2H_6+C_4H_6+CO_2$

(D) $C_2H_6 + C_4H_{10}+CO_2\uparrow$

Q8. Electrolysis and decarboxylation of $CH_3-\underset{\underset{CH_3}{|}}{CH}COONa$ respectively give :-

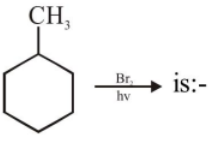
(A) 2,3-dimethylbutane and propane

(B) Propane and 2,3-dimethylbutane

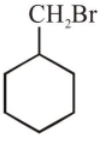
(C) Isobutane and propane

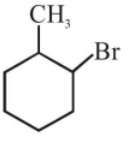
(D) n-butane and propane

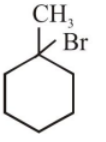
Q9. The major product obtained in the reaction,

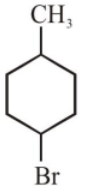


is:-

(A) 

(B) 

(C) 

(D) 

Q10. Nitration of propane with nitric acid vapour at about 430°C is expected to yield :-

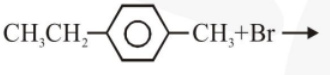
(A) 1-nitropropane

(B) 2-nitropropane

(C) Nitroethane and Nitromethane

(D) A mixture of all the above

Q11. Consider the following reaction :



Which of the following types of hydrogen atoms are most easily abstracted by the bromine radical to form HBr ?

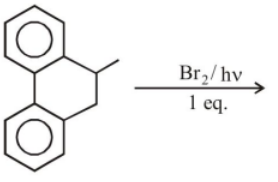
(A) Hydrogen atoms attached to a benzene ring.

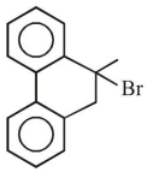
(B) Hydrogen atoms of the methyl group attached directly to benzene ring.

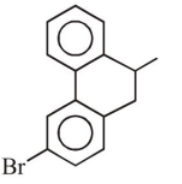
(C) Hydrogen atoms of the methylene group of the ethyl substitutent.

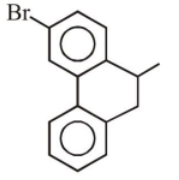
(D) None of these

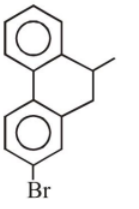
Q12. The most appropriate product.

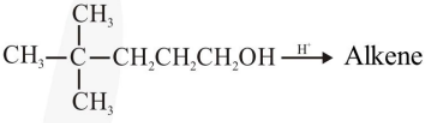


(A) 

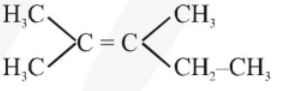
(B) 

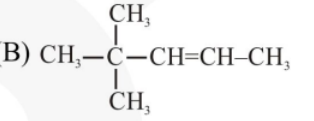
(C) 

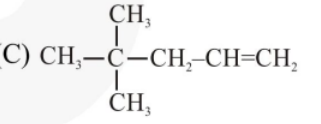
(D) 

Q13. 

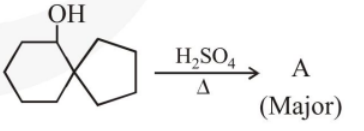
The major alkene formed in the above reaction will be:-

(A) 

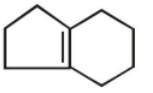
(B) 

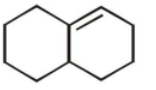
(C) 

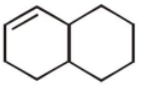
(D) none of the above

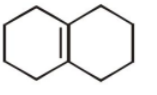
Q14. 

Major product A is :

(A) 

(B) 

(C) 

(D) 

Q15. $CH_3–CH_2–CH_2–CH_2–OH \xrightarrow[\Delta]{\text{Conc. } H_2SO_4}$

Major Product (Y) is :-

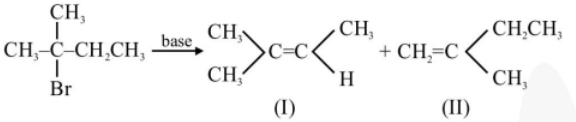
(A) cis-2-butene

(B) trans-2-butene

(C) 1-butene

(D) Iso-butene

Q16. Consider the following reaction :



(I) (II)

Which of the following base will give the best yield of the alkene II as the major product :-

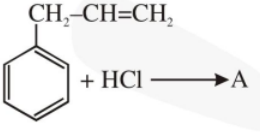
(A) CH_3O^-

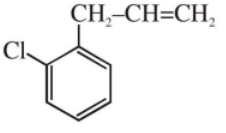
(B) $C_2H_5O^-$

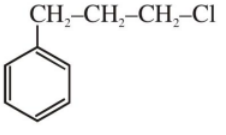
(C) $(CH_3)_3CO^-$

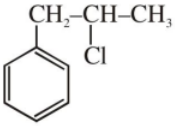
(D) $(C_2H_5)_3CO^-$

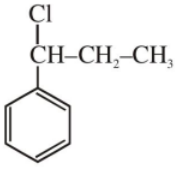
Q17. What is 'A' in the following reaction :



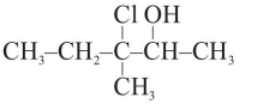
(A) 

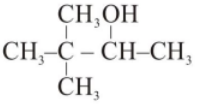
(B) 

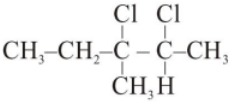
(C) 

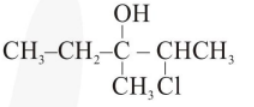
(D) 

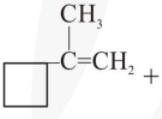
Q18. 3–Methyl–2–pentene on reaction with HOCl gives-

(A) 

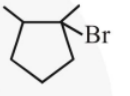
(B) 

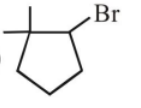
(C) 

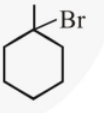
(D) 

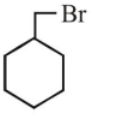
Q19. 

Product (A) is :-

(A) 

(B) 

(C) 

(D) 

Q20. Propanal and pentan -3-one are ozonolysis product of –

(A) $CH_3-CH_2-CH=\underset{\displaystyle CH_2-CH_3}{\overset{\displaystyle |}{C}}-CH_2-CH_3$

(B) $CH_3-CH=\underset{\displaystyle CH_2-CH_3}{\overset{\displaystyle |}{C}}-CH_2-CH_3$

(C) $CH_3-CH_2-\underset{\displaystyle CH_3-C-CH_3}{\overset{\displaystyle \|}{C}}-CH_3$

(D) $CH_3-CH=CH-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-CH_2-CH_2-CH_3$

Q21. $CH_3-CH_2-\underset{Cl}{CH}-CH_3 \xrightarrow[\Delta]{alc.KOH} X \xrightarrow[Zn/H_2O]{O_3} (y)$

Product (y) is -

(A) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (2 mole)

(B) $CH_3-\overset{O}{\overset{\|}{C}}-H$ (2 mole)

(C) $H-\overset{O}{\overset{\|}{C}}-H$ (2 mole)

(D) $CH_3-CH=CH-CH_3$

Q22. $X \xrightarrow[Zn]{O_3} \underset{CHO}{\overset{CHO}{|}} CH$ (2 mole) ; (X) is -

(A) Aromatic compound

(B) Non-aromatic compound

(C) Anti-aromatic compound

(D) All of these

Q23. $CH_3-\underset{CH_3}{\overset{}{C}}=CH-C_6H_5$ $\xrightarrow{HBr/\text{Peroxide}}$ 'A'; $\xrightarrow{HBr}$ 'B'

Product 'A' & 'B' are respectively–

(A) $CH_3-\underset{CH_3}{\overset{Br}{C}}-CH_2-C_6H_5$ &

$CH_3-\underset{CH_3}{CH}-\underset{Br}{CH}-C_6H_5$

(B) $CH_3-\underset{CH_3}{CH}-\underset{Br}{CH}-C_6H_5$ &

$CH_3-\underset{CH_3}{\overset{Br}{C}}-CH_2-C_6H_5$

(C) Both are $CH_3-\underset{CH_3}{\overset{Br}{C}}-CH_2-C_6H_5$

(D) Both are $CH_3-\underset{CH_3}{CH}-\overset{Br}{CH}-C_6H_5$

Q24. Acetylene is obtained by the electrolysis of:-

(A) Sodium succinate

(B) Potassium fumarate

(C) Both (A) and (B)

(D) None of these

Q25. Consider the following reaction sequence

$CH_3-C\equiv CH \xrightarrow[2\,mol]{HCl} A \xrightarrow[heat]{aq.KOH} B$

The product (A) and (B) are, respectively :-

(A) $CH_3CH_2CHCl_2$ and CH_3CH_2CHO

(B) $CH_3CCl_2CH_3$ and $CH_3-C\equiv CH$

(C) $CH_3CCl_2CH_3$ and CH_3CO-CH_3

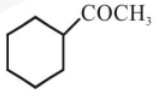
(D) $CH_3-CHCl-CH_2Cl$ and $CH_3CHOHCH_2OH$

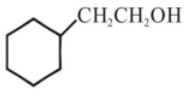
Q26. 1-Butyne and 2-Butyne can not be distinguished by reaction with the following except :-

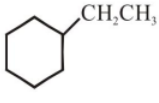
(A) $[Ag(NH_3)_2]^+$ (B) $[Cu(NH_3)_2]^+$

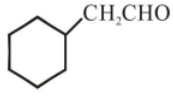
(C) Baeyer's reagent (D) Sodamide

Q27. Hydration of  in presence of $H_2SO_4/HgSO_4$ gives :-

(A) 

(B) 

(C) 

(D) 

Q28. Propyne on polymerisation gives :-

(A) Mesitylene (B) Benzene

(C) Ethyl benzene (D) Propyl benzene

Q29. When phenyl magnesium bromide reacts with t-butanol, the product would be :-

(A) Benzene

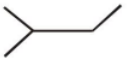
(B) Phenol

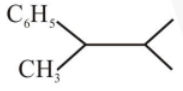
(C) t-butyl benzene

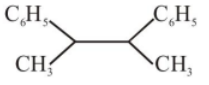
(D) t-butyl phenyl ether

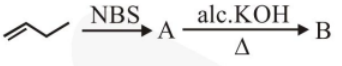
Q30. $CH_3-CH(CH_3)-MgCl \xrightarrow{C_6H_5OH} A \xrightarrow{Cl_2,h\nu} B \xrightarrow[\text{Dry ether}]{Na} C$, what is C ?

(A) 

(B) 

(C) 

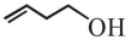
(D) 

Q31. 

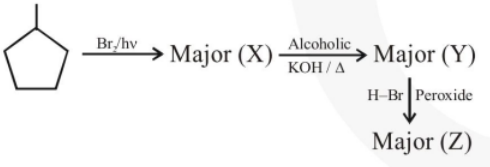
Identify 'B' product :-

(A) 

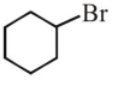
(B) 

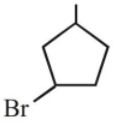
(C) 

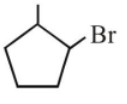
(D) 

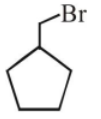
Q32. 

Product (Z) is :-

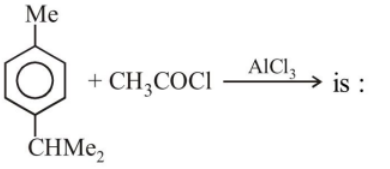
(A) 

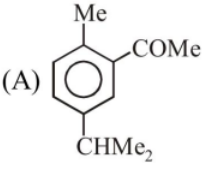
(B) 

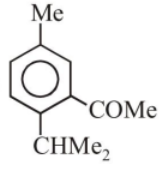
(C) 

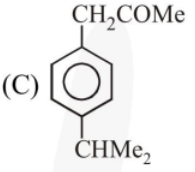
(D) 

Q33. The major product of the reaction,



(A) 

(B) 

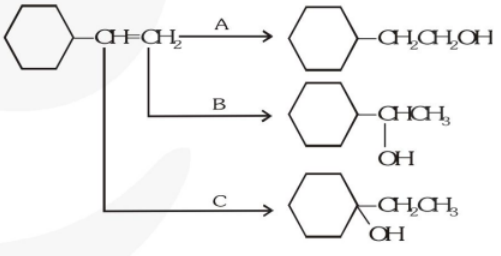
(C) 

(D) None of these

Q34. Compound 'A' on chlorination gives compound 'B', compound 'B' reacts with alc. KOH gives gas 'C', which decolourise Baeyer reagent. ozonolysis of compound 'C' gives only HCHO compound. 'A' is :

(A) C_2H_6 (B) C_2H_4

(C) C_4H_{10} (D) C_2H_5Cl

Q35. 

A, B and C are :

(A) simple hydration

(B) hydroboronation, mercuration-demercuration, hydration

(C) hydration, hydroboronation, mercuration-demercuration

(D) mercuration-demercuration, hydration, hydroboronation

Exercise - 2A Brain Booster

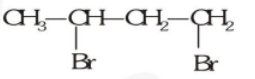
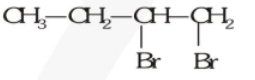
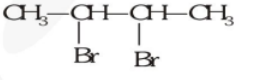
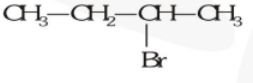
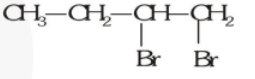
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE OR MORE THAN** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 1 to 3

$X(C_4H_9Br) \xrightarrow{alc.KOH} Y \xrightarrow{Br_2} Z\ (Dibromide) \xrightarrow{NaNH_2} W\ (gas) \xrightarrow[Sol^n]{AgNO_3}$ white ppt. X, Y and Z are:

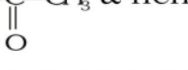
Q1.

	X	Y	Z
(A)	$CH_3—CH_2CH_2CH_2Br$	$CH_3—CH=CH—CH_3$	
(B)	$CH_3—CH_2CH_2CH_2Br$	$CH_3—CH_2—CH=CH_2$	
(C)	$CH_3—CH_2—CH_2—CH_3$	$CH_3—CH=CH—CH_3$	
(D)		$CH_3—CH=CH—CH_3$	

Q2. Reductive ozonolysis of Y yields :

(A) 2 moles of CH_3CHO

(B) CH_3CH_2CHO & HCHO

(C)  & HCHO

(D) CH_3CH_2COOH & HCOOH

Q3. Which of the following statement (s) is/are correct :

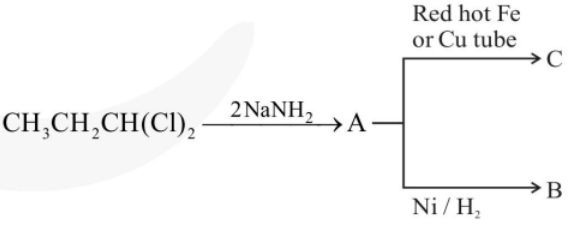
(A) Compound W has 2 DU

(B) Y & W are functional isomers

(C) W can be converted into Y with Lindlar catalyst

(D) W can be converted into Y with Ni/Pt

Paragraph for Q. 4 to 5



4. A is -

(A) Propyne (B) Propene

(C) Propanal (D) Propanone

5. C is -

(A) Mesitylene (B)Benzene

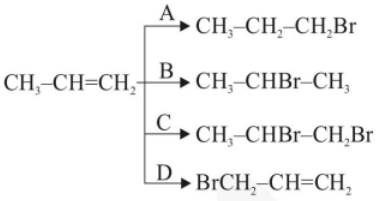
(C) Cyclooctatetraene (D)Benzaldehyde

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q6. Match the List :

List-I



List-II

(i) HBr

(ii) HBr + peroxide

(iii) NBS

(iv) Br_2, low temperature

	A	B	C	D
(A)	(ii)	(i)	(iv)	(iii)
(B)	(i)	(ii)	(iv)	(iii)
(C)	(i)	(iii)	(iv)	(ii)
(D)	(ii)	(i)	(iii)	(iv)

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

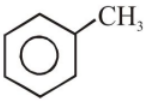
Q7. Match the column I with column II.

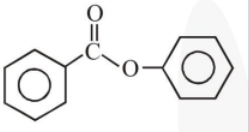
Column-I	**Column-II**
(A) Dehydrohalogenation of alkyl halides	(p) Kolbe reaction
(B) Electrolysis of sodium salt	(q) Alc. KOH
(C) Ozonolysis	(r) Addition product of ethylene
(D) Dichloro ethylene	(s) Sodalime
(E) Decarboxylation	(t) Alkene

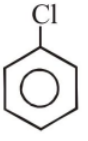
NUMERICAL TYPE

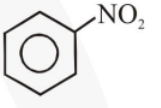
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

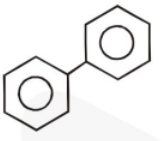
Q8. How many of the following compounds will show higher rate of electrophilic substitution reaction than benzene -

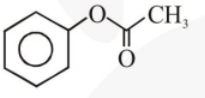
(a) 

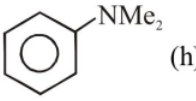
(b) 

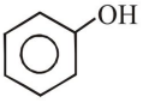
(c) 

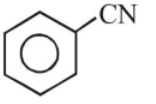
(d) 

(e) 

(f) 

(g) 

(h) 

(i) 

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q9. **Statement-I :** 1-Butene on reaction with HBr in the presence of a peroxide produces 1-bromobutane.

Statement-II : It involves the formation of a primary radical.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	A	C	B	A	B	A	B	A	C
11	12	13	14	15	16	17	18	19	20
A	A	B	D	D	A	D	C	B	C
21	22	23	24	25	26	27	28	29	30
A	A	B	C	A	C	C	A	C	B
31	32	33	34	35	36	37	38	39	40
C	D	B	B	A	A	B	A	B	C

Ex - 1A

1. A, C, D **2.** A, B, C **3.** (A) → q,s ; B → r ; (C) → q ; (D) → q

4. 003 **5.** A

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	D	A	C	A	D	A	C	D
11	12	13	14	15	16	17	18	19	20
C	A	B	D	B	D	D	D	A	A
21	22	23	24	25	26	27	28	29	30
B	C	A	B	C	C	A	A	A	A
31	32	33	34	35					
B	C	A	A	B					

Ex - 2A

1. B **2.** B **3.** A, C **4.** A **5.** A

6. A **7.** (A) → q ; (B) → p ; (C) → t ; (D) → r ; (E) → s

8. 5 **9.** C

-: Important Notes & Formulas :-